

Lab 3: OS Installation – Implementation Sequence

Implementation Sequence

The following steps describe the complete process of installing VirtualBox and Ubuntu OS on a Windows host machine, as performed during the lab session.

Step 1: Download VirtualBox Installer

Open the official VirtualBox website (www.virtualbox.org/wiki/Downloads) and download the installer for your host operating system. In this case, the Windows host package was selected and downloaded.

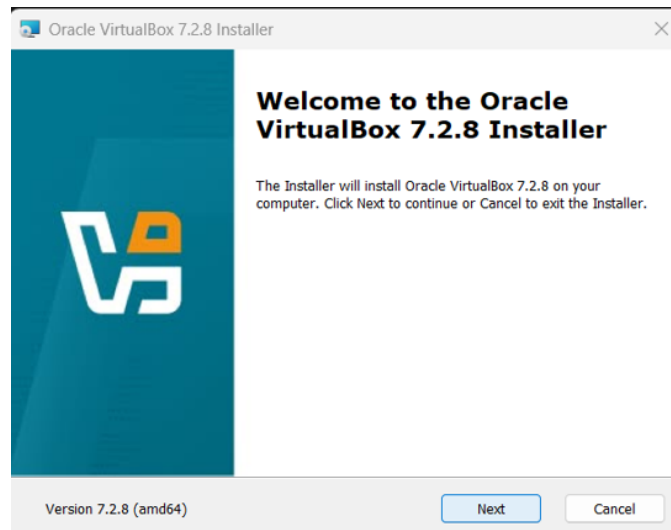


Figure 1: VirtualBox official download page

Step 2: Launch the VirtualBox Setup Wizard

After downloading, run the VirtualBox installer (.exe). The Setup Wizard welcome screen appears. Click 'Next' to proceed with the installation.

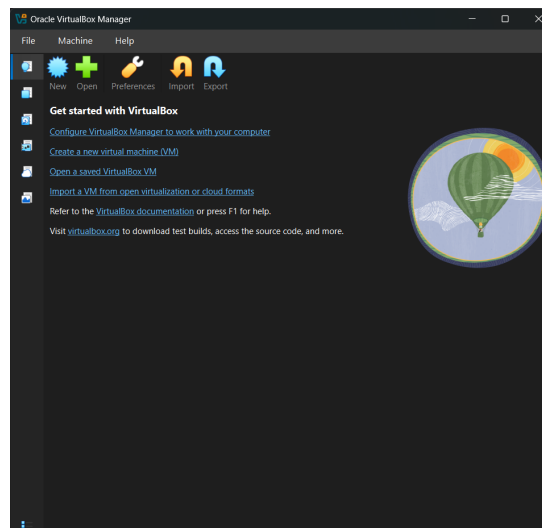


Figure 2: VirtualBox Setup Wizard – Welcome screen

Step 3: Select Installation Location and Features

The wizard allows selection of components to install and the destination folder. Leave the defaults selected to install all core components including VirtualBox Application, USB support, and Networking features. Click 'Next' to continue.

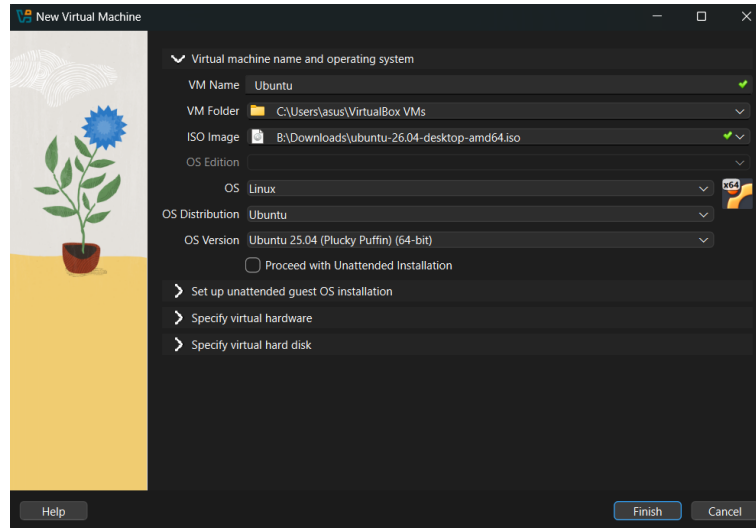


Figure 3: Selecting installation location and features

Step 4: Network Interface Warning

A warning dialog appears indicating that VirtualBox will temporarily reset the network interface during installation. This is normal behavior. Click 'Yes' to allow and proceed.

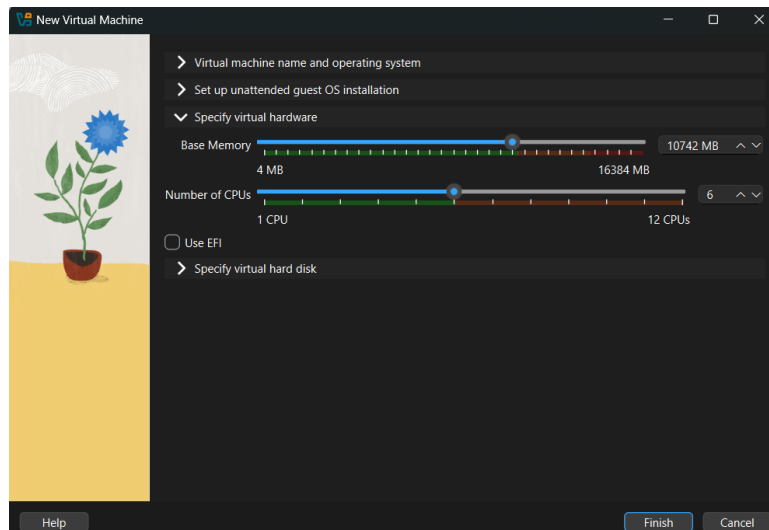


Figure 4: Network interface warning dialog

Step 5: Missing Dependencies Warning

VirtualBox may prompt about missing Python dependencies or other optional components. Acknowledge the warning and click 'Yes' to continue without optional dependencies, or install them if required.

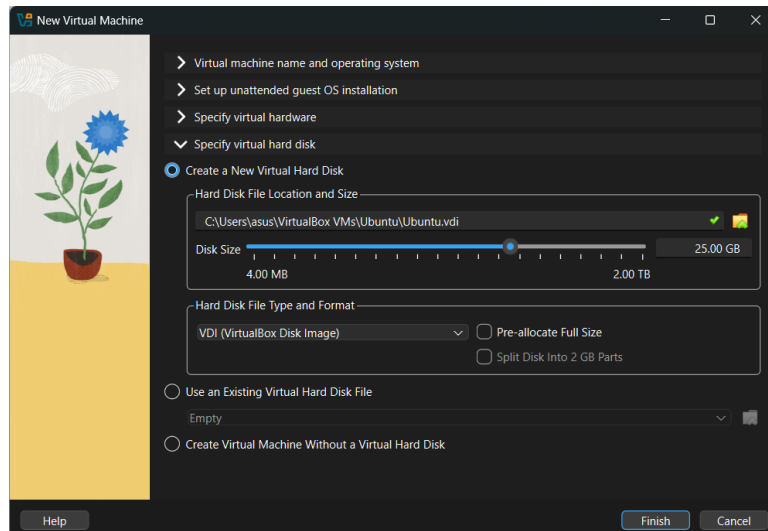


Figure 5: Missing dependencies warning

Step 6: Ready to Install

The wizard is now ready to begin the installation. A summary of the selected options is shown. Click 'Install' to start copying files and installing VirtualBox on the system.

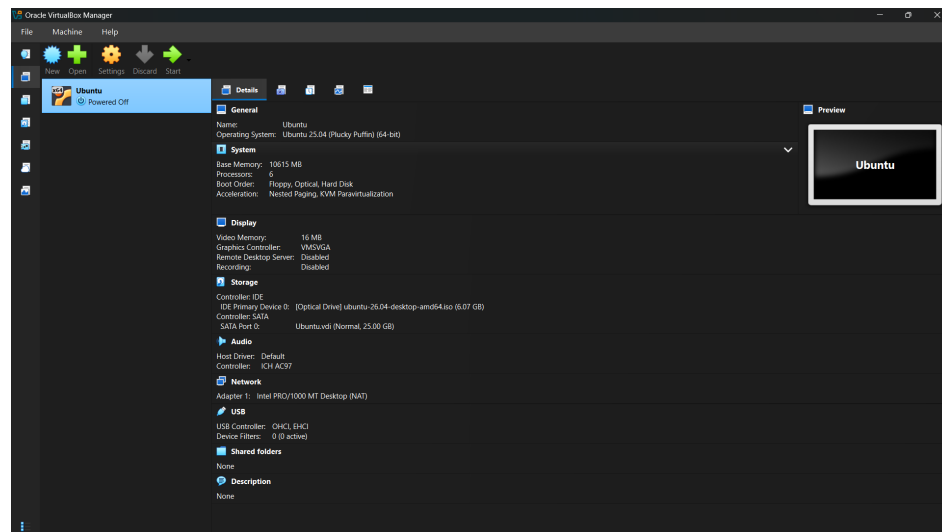


Figure 6: Ready to Install screen – click Install to begin

Step 7: Installation Progress

VirtualBox installs all required components including device drivers and virtual network adapters. A progress bar indicates the installation status. Wait for it to complete.

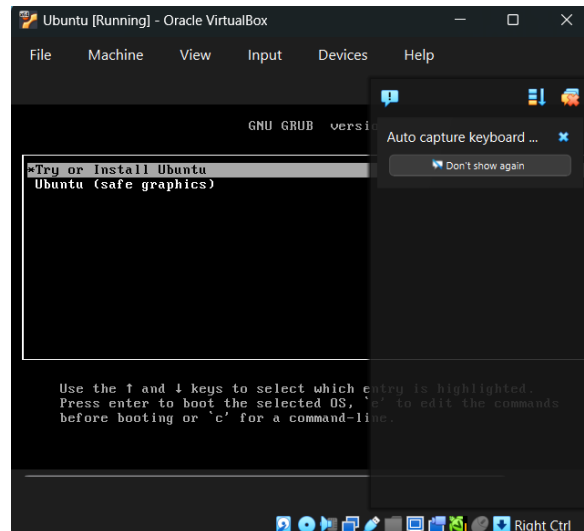


Figure 7: Installation in progress

Step 8: VirtualBox Installation Complete

The installation completes successfully. The final screen confirms VirtualBox has been installed. Ensure 'Start Oracle VirtualBox after installation' is checked, then click 'Finish'.

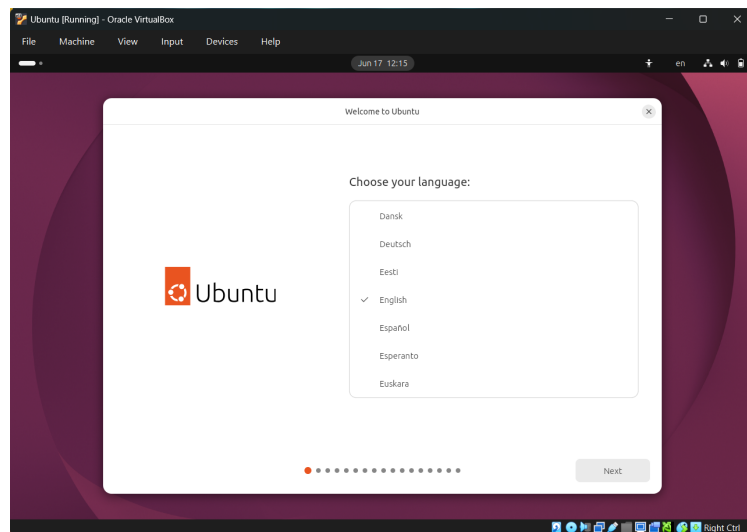


Figure 8: Installation complete – VirtualBox successfully installed

Step 9: VirtualBox Manager Opens

VirtualBox Manager launches. This is the main interface for creating and managing virtual machines. Currently no VMs exist. Click 'New' to create a new virtual machine for Ubuntu.

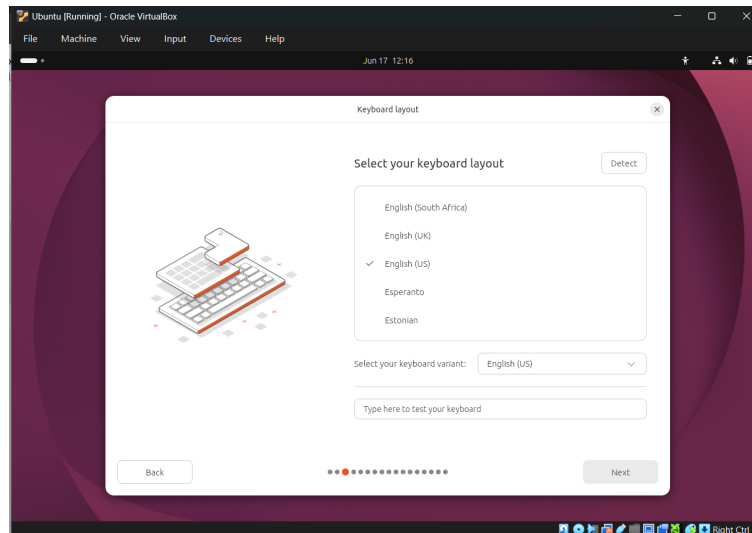


Figure 9: VirtualBox Manager – main interface

Step 10: Create New Virtual Machine – Name and OS Type

In the 'Create Virtual Machine' dialog, enter a name for the VM (e.g., 'Ubuntu'). Set the Type to 'Linux' and Version to 'Ubuntu (64-bit)'. Select the folder where the VM will be stored. Click 'Next' to continue.

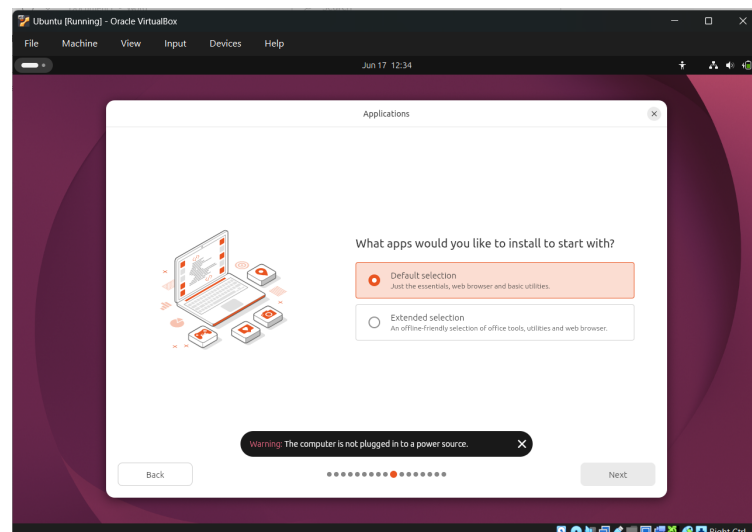


Figure 10: Setting VM name, type, and version

Step 11: Allocate Memory (RAM)

Set the amount of RAM to allocate to the virtual machine. A minimum of 2048 MB (2 GB) is recommended for Ubuntu. Allocating 4096 MB (4 GB) provides better performance. Adjust using the slider and click 'Next'.

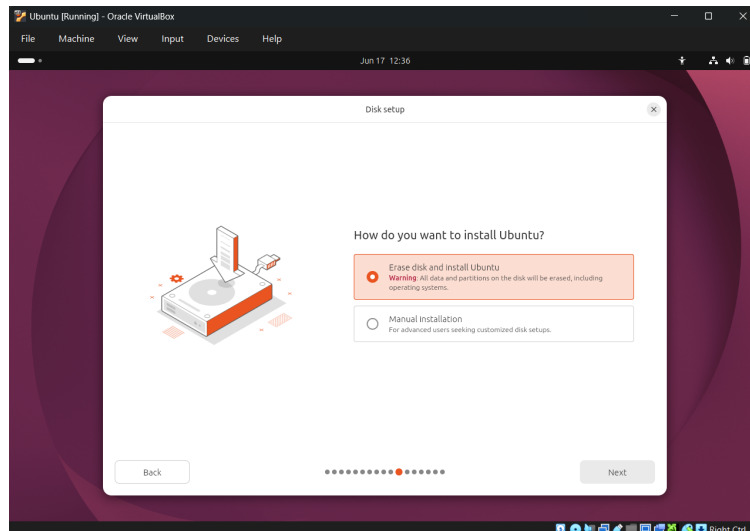


Figure 11: Allocating RAM to the virtual machine

Step 12: Create a Virtual Hard Disk

Select 'Create a virtual hard disk now' and click 'Create'. This will create a virtual storage drive for the Ubuntu OS to be installed on. Choose the VDI (VirtualBox Disk Image) format and allocate at least 20 GB of disk space.

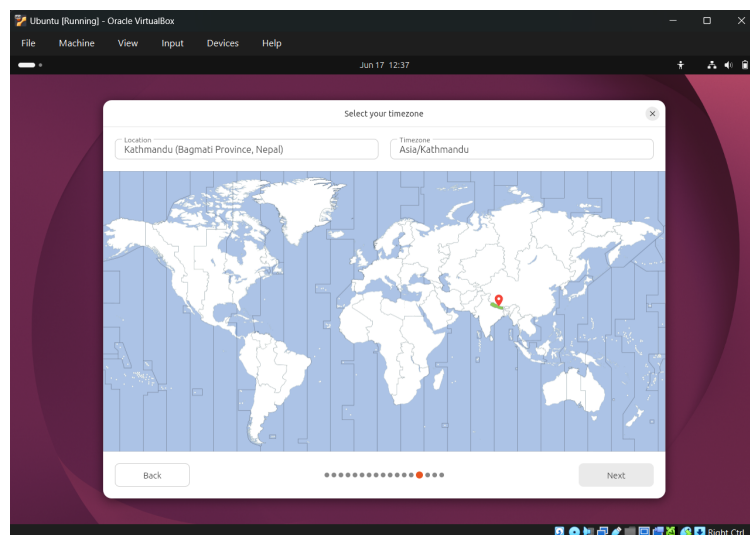


Figure 12: Virtual hard disk creation settings

Step 13: Load Ubuntu ISO and Start VM

Before starting the VM, attach the Ubuntu ISO file as a virtual optical drive. Go to VM Settings → Storage → Controller: IDE → Empty → Choose Disk File, and select the downloaded Ubuntu ISO. Then start the VM by clicking 'Start'.

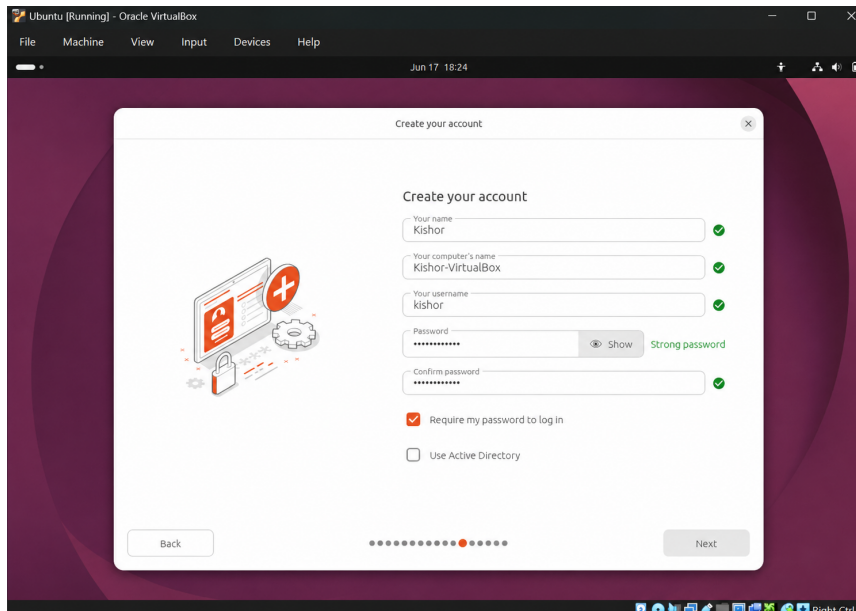


Figure 13: Ubuntu ISO loaded and VM ready to start

Step 14: Ubuntu Boot Menu

The VM boots from the Ubuntu ISO. The GRUB boot menu appears. Select 'Try or Install Ubuntu' using the keyboard arrow keys and press Enter to proceed with the Ubuntu installation.

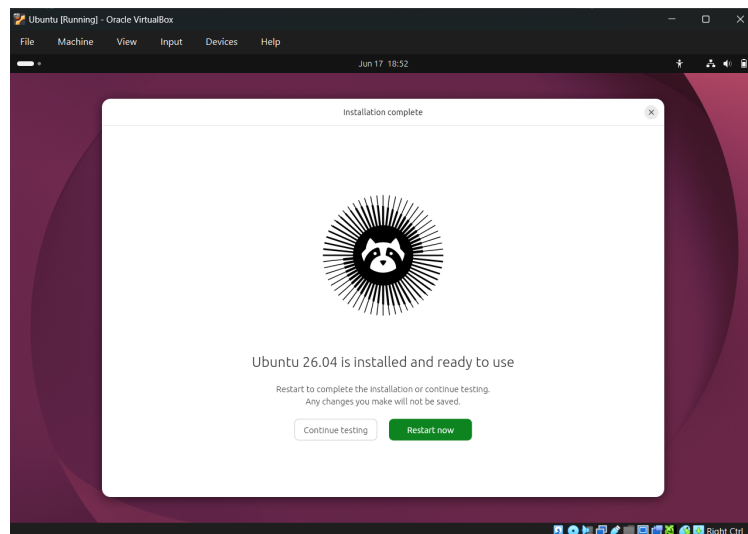


Figure 14: Ubuntu GRUB boot menu

Step 15: Ubuntu Installation – Language and Options

The Ubuntu installer loads. Select the preferred language (English) and choose 'Install Ubuntu'. Select keyboard layout and proceed to installation type. Choose 'Erase disk and install Ubuntu' for a clean virtual disk installation.

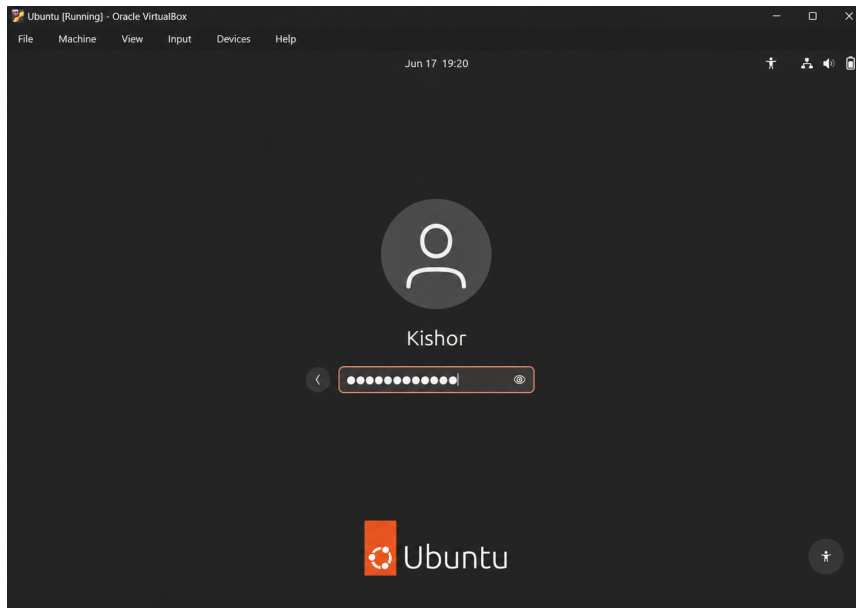


Figure 15: Ubuntu installer – language selection and install options

Step 16: Ubuntu Installation Complete – Desktop Ready

Ubuntu finishes installing and restarts the VM. After removing the virtual ISO, the system boots into the Ubuntu desktop environment (GNOME). The installation is complete and Ubuntu OS is now running inside VirtualBox.

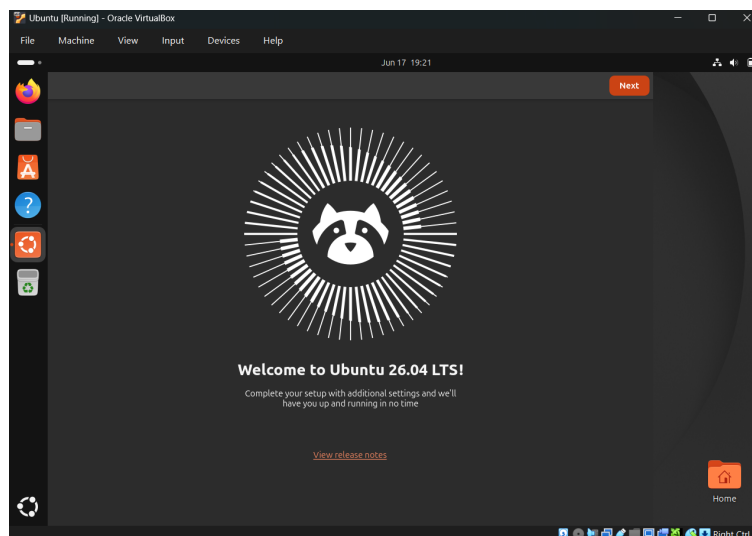


Figure 16: Ubuntu desktop – installation successful